

$$\ddot{U} + U(\theta) - \frac{1}{a} - \int U^2(\theta) = 0$$

$$\ddot{U}(\theta) = \int U^2(\theta) + \frac{1}{a} - U(\theta)$$

$$\begin{cases} \dot{U}(\theta) = V(\theta) \\ \dot{V}(\theta) = \int U^2(\theta) + \frac{1}{a} - U(\theta) \end{cases}$$

$$\frac{d}{d\theta} (U, V) = \left(V, \int U^2(\theta) + \frac{1}{a} - U(\theta) \right)$$

$$\underline{\int_{-1}^1}$$

$$(\dot{U}, \dot{V}) = \left(V, -U(\theta) + \frac{1}{a} \right)$$

$$\ddot{u} + u = \frac{1}{2} \quad u = e^{\lambda \theta} \therefore$$

$$\lambda^2 e^{\lambda \theta} + \lambda e^{\lambda \theta} = \frac{1}{2}$$

$$\therefore \text{Sol gen} = \lambda^2 \cdot e^{\lambda \theta} + e^{\lambda \theta} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\lambda = \pm i$$

$$\lambda = 0 \pm 1 \cdot i$$

$$\therefore u(\theta) = e^{0 \cdot \theta} [A \cdot \cos(1 \cdot \theta) + B \sin(1 \cdot \theta)]$$

$$u(\theta) = A \cdot \cos(\theta) + B \cdot \sin(\theta)$$

$$u_H + u_P = \frac{1}{2} \quad \text{can } u_P = C \text{ etc}$$

$$u_H = 0$$

$\therefore$

$$0 + C = \frac{1}{2} \Rightarrow u(\theta) = u_H + \frac{1}{2}$$

∴

$$V(\theta) = A \cdot \cos(\theta) + B \cdot \sin(\theta) + \frac{1}{a}$$

$$V(\theta) = \frac{1}{V(\theta)}$$

∴

$$V(0) = \frac{a}{1+\epsilon} = \frac{1}{\underbrace{A \cdot \cos(0)}_A + \underbrace{B \cdot \sin(0)}_0 + \frac{1}{a}}$$

$$\frac{a}{1+\epsilon} = \frac{1}{A + \frac{1}{a}}$$

$$a \left(A + \frac{1}{a} \right) = 1 + \epsilon$$

$$a \cdot A + 1 = 1 + \epsilon$$

$$a \cdot A = \epsilon$$

$$A = \frac{\epsilon}{a}$$

$$\dot{r} = \frac{dr}{d\theta} \cdot \dot{\theta}$$

$$r(\theta) = r(u(\theta)^{-1})$$

$\therefore$

$$\dot{r}(\theta) = \frac{dr}{d\theta} = \frac{dr}{du} \cdot \frac{du}{d\theta}$$

$\downarrow$
 $-\frac{1}{u^2}$

$$0 = \frac{dr}{d\theta} = -\frac{1}{u^2} \cdot \frac{du}{d\theta}$$

$$0 = \frac{du}{d\theta}(\theta) = \dot{u}(\theta)$$

$\therefore$

$$\dot{u}(\theta) = -A \cdot \sin(\theta) + B \cdot \cos(\theta)$$

$$\dot{u}(0) = 0 + B = 0 \Rightarrow B = 0$$

$$\therefore V(\theta) = \frac{\varepsilon \cdot \cos(\theta)}{\alpha} + \frac{1}{\alpha} = \frac{1}{\alpha} (\varepsilon \cdot \cos(\theta) + 1)$$

$$\therefore r(\theta) = \frac{\alpha}{(\varepsilon \cdot \cos(\theta) + 1)}$$

2] $\alpha = 1$

$$\Rightarrow r(\theta) = \frac{1}{\varepsilon \cos(\theta) + 1}$$

$$\therefore \varepsilon \cdot \cos(\theta) = -1 \rightarrow \frac{1}{0} \text{ (AB)}!$$

$$\cos(\theta) = -\frac{1}{\varepsilon}$$

$$\theta = \arccos\left(-\frac{1}{\varepsilon}\right)$$

$$1 \leq -\frac{1}{\varepsilon} \leq 1$$

$$\left| \frac{1}{\varepsilon} \right| \leq 1 \rightarrow \varepsilon \geq 1$$

" El denominador puede hacerse 0 cuando


$$\ell = 1 \text{ (parabola)} \Rightarrow 1 + \cos \theta = 0 \Rightarrow \cos(\theta) = -1$$

$$\Rightarrow \theta = \pi \Rightarrow \text{den} = 0 \text{ (orbiten abbrechen)}$$
$$|\varepsilon| > 1 \text{ (hyperbolic)} \Rightarrow \exists \theta / \cos(\theta) = -\frac{1}{\varepsilon} \Rightarrow r \rightarrow \infty$$

Naturaleza de las Raíces	Tipo de Autovalores (λ)	Solución General $u(\theta)$	Comportamiento Típico
Reales y Distintas	$\lambda_1 \neq \lambda_2$	$u(\theta) = C_1 e^{\lambda_1 \theta} + C_2 e^{\lambda_2 \theta}$	Crecimiento o decaimiento exponencial puro.
Reales y Repetidas	$\lambda_1 = \lambda_2 = \lambda$	$u(\theta) = C_1 e^{\lambda \theta} + C_2 \theta e^{\lambda \theta}$	Aparece el término para asegurar la independencia lineal.
Complejas Conjugadas	$\lambda = \alpha \pm \beta i$	$u(\theta) = e^{\alpha \theta} [C_1 \cos(\beta \theta) + C_2 \sin(\beta \theta)]$	Oscilaciones (si $\alpha < 0$ son amortiguadas, si $\alpha > 0$ son crecientes).

$$3 \left\{ \begin{array}{l} \dot{v}(\theta) = r(\theta) \\ \dot{r}(\theta) = 0,05 v^2(\theta) + \frac{1}{a} - v(\theta) \end{array} \right.$$

$$\left\{ \begin{array}{l} \dot{v}(\theta) = r(\theta) \\ \dot{r}(\theta) = 0,05 v^2(\theta) + \frac{1}{a} - v(\theta) \end{array} \right.$$

$$\ddot{v} + v - 0,05 v^2 = \frac{1}{a}$$

44/

$$\lambda^2 \cdot e^{\lambda \cdot \theta} + e^{\lambda \cdot \theta} - 0,05 e^{2\lambda \cdot \theta} = 0$$

$$\lambda^2 + 1 - 0,05 = 0$$

$$\lambda^2 = -0,95$$

$$\lambda = \pm i \cdot \sqrt{0,95}$$

$$\therefore v_H = e^{i\theta} \left[A \cos(\sqrt{0.95} \cdot \theta) + B \sin(\sqrt{0.95} \cdot \theta) \right]$$

$$v_H = A \cos(\sqrt{0.95} \cdot \theta) + B \sin(\sqrt{0.95} \cdot \theta)$$

~~v_P~~

$$v_P = C \text{ (cte)}$$

$$\therefore v_H + v_P = \frac{1}{\omega} \Rightarrow 0 + C = \frac{1}{\omega} \Rightarrow C = \frac{1}{\omega}$$

$\therefore$

Sol ~~for~~
$$v(\theta) = A \cos(\sqrt{0.95} \cdot \theta) + B \sin(\sqrt{0.95} \cdot \theta) + \frac{1}{\omega}$$

$$V(\theta) = \frac{a}{1+\epsilon}$$

$\therefore$

$$V(\theta) = \frac{1+\epsilon}{a} = A \cdot \cos(\theta) + \underbrace{B \cdot \sin(\theta)}_0 + \frac{1}{a}$$

$$\frac{1+\epsilon}{a} = A + \frac{1}{a}$$

$$\cancel{1+\epsilon} = a \cdot A \cancel{+ 1}$$

$$\boxed{\frac{\epsilon}{a} = -A}$$

$$\dot{r} = \frac{dr}{d\theta} = \frac{dr}{dv} \cdot \frac{dv}{d\theta} = 0$$

$$= -\frac{1}{v^2} \cdot \frac{dv}{d\theta} = 0$$

$$\therefore \frac{dv}{d\theta}(\theta) = 0$$

$$\dot{v}(\theta) = 0$$

∴

$$\dot{v}(\theta) = -A \cdot \sin(\sqrt{0.95} \theta) + B \cdot \cos(\sqrt{0.95} \theta)$$

$$\dot{v}(0) = 0 + B = 0 \rightarrow B = 0$$

∴

$$v(\theta) = \frac{a}{\varepsilon \cdot \cos(\theta) + 1} \Rightarrow a = 1 \Rightarrow \frac{1}{\varepsilon \cdot \cos(\theta) + 1}$$

Los gráficos son los mismos que en (2) $\left\{ \begin{array}{l} 0 \leq \varepsilon < 1 \\ \varepsilon = 1 \\ \varepsilon > 1 \end{array} \right.$